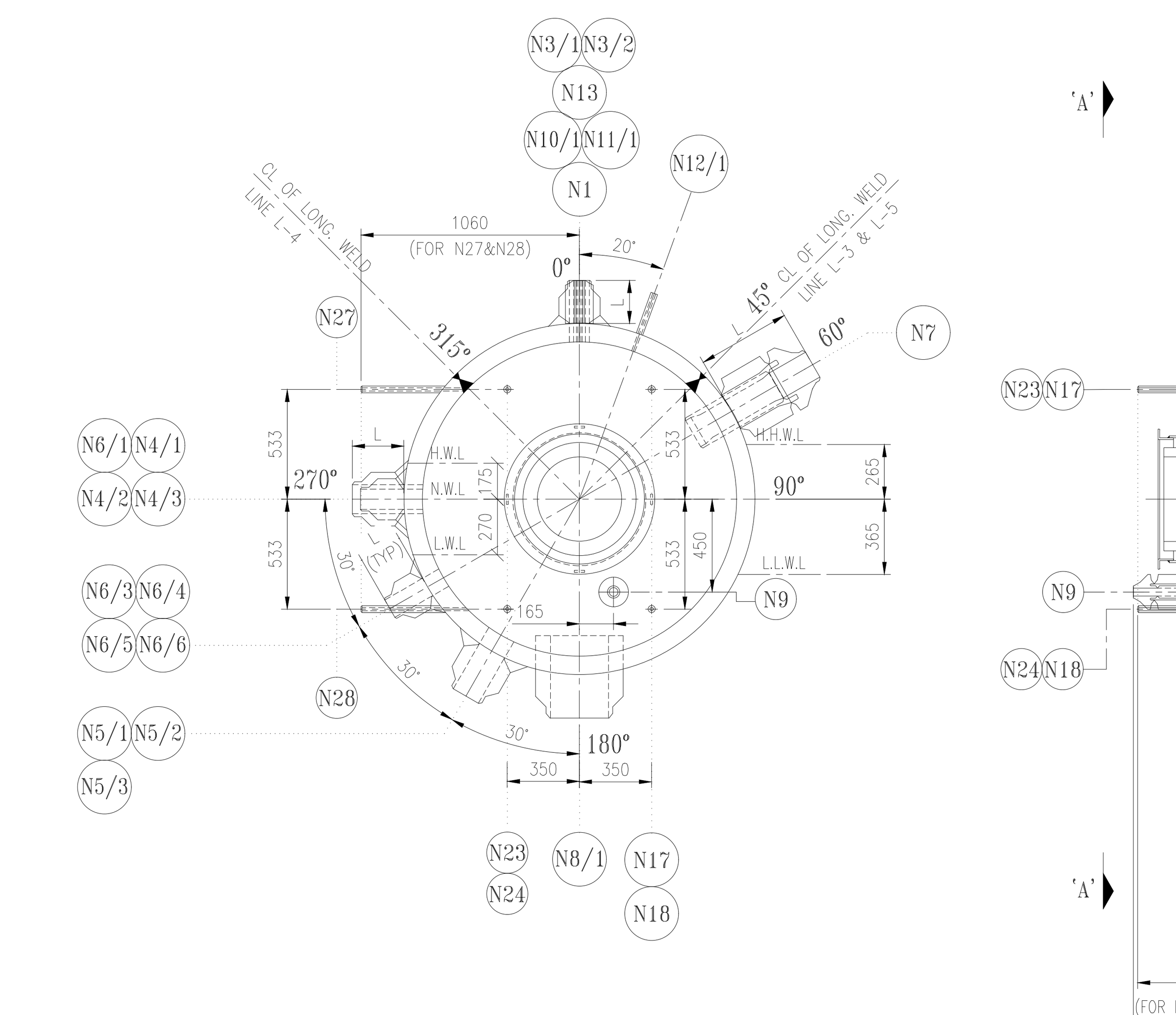
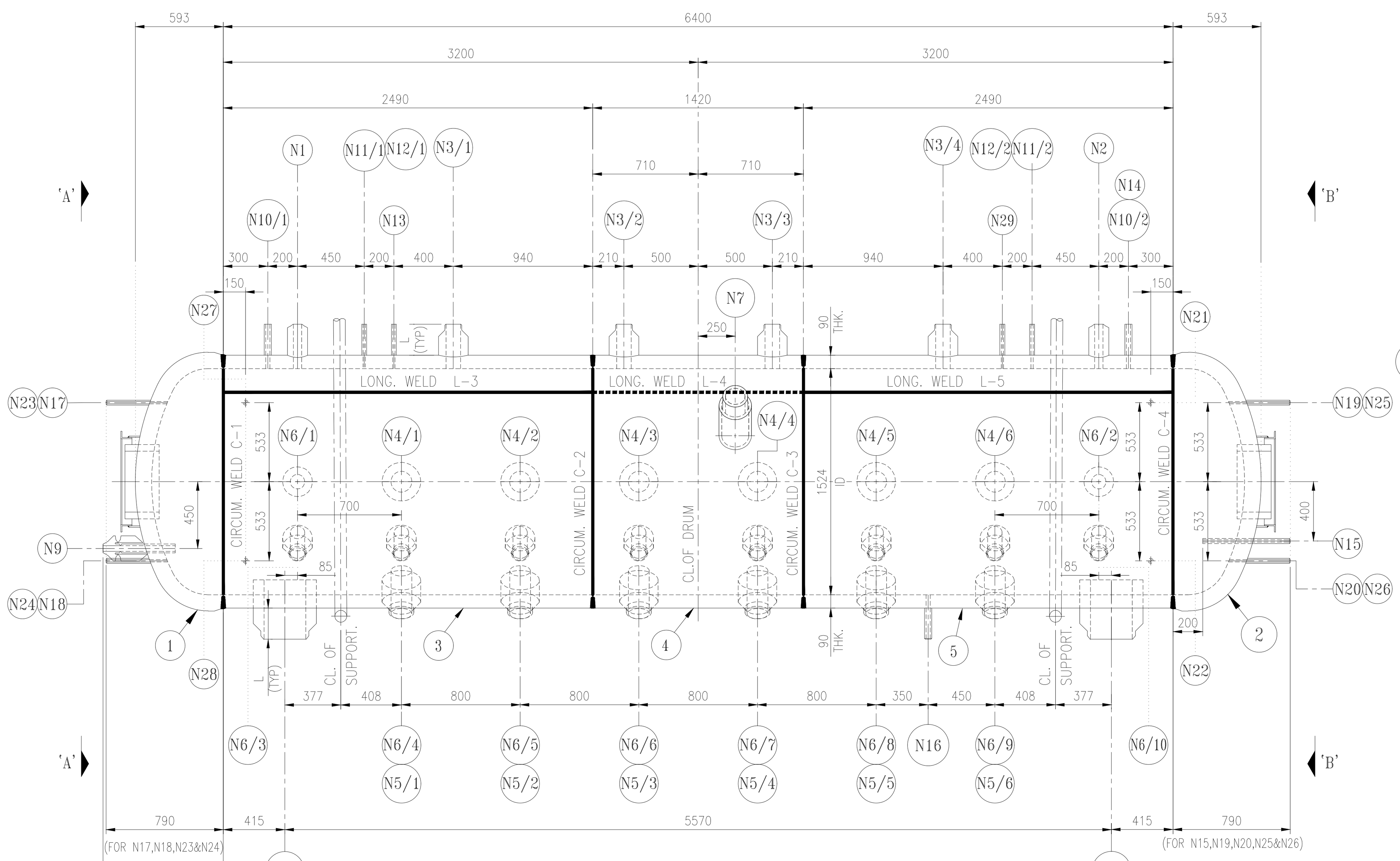
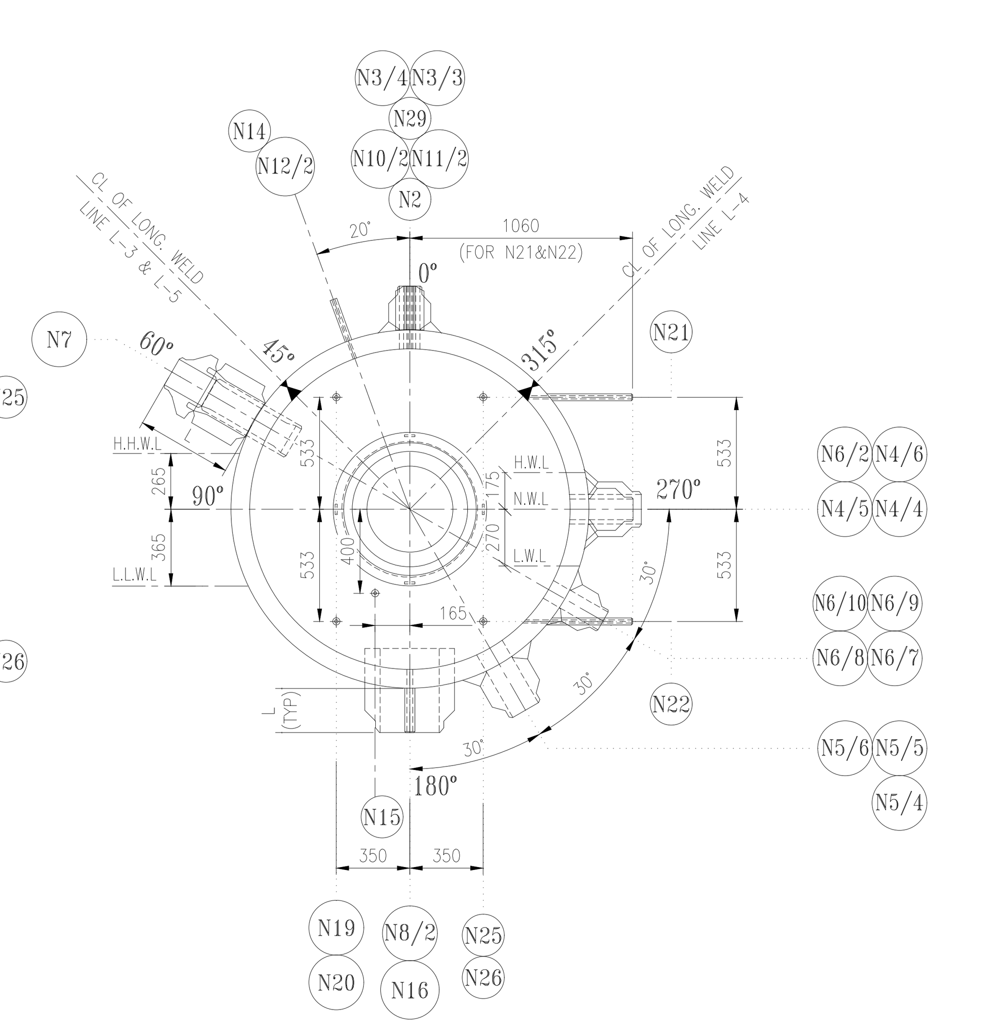
PROJECT ID JB1171
DOCUMENT ID JB1171-50010100-007-0001



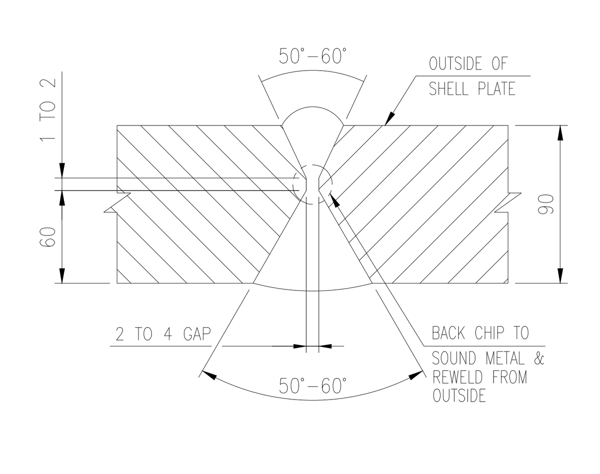
VIEW -'A' 'A'
VIEW FROM LHS OF BOILER



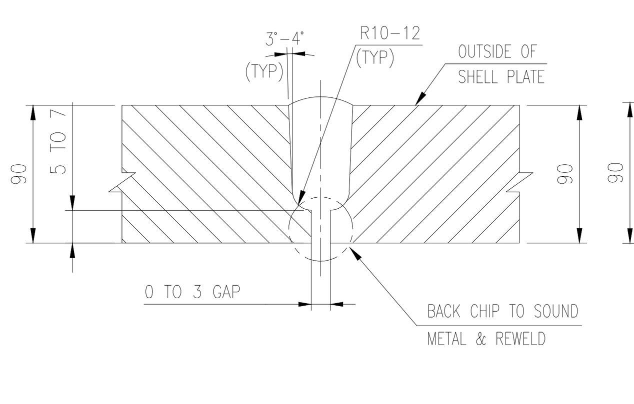
ARRANGEMENT OF STEAM DRUM
VIEW FROM FRONT END OF BOILER



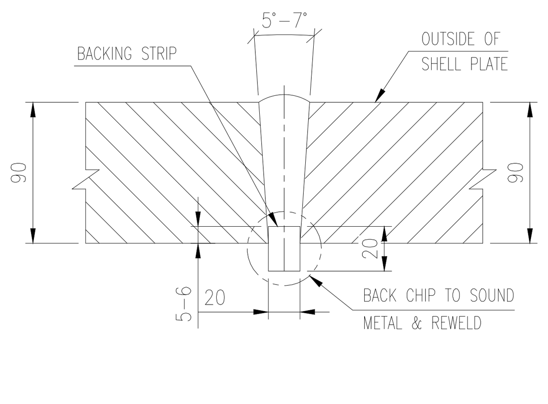
VIEW -'B' 'B'
VIEW FROM RHS OF BOILER



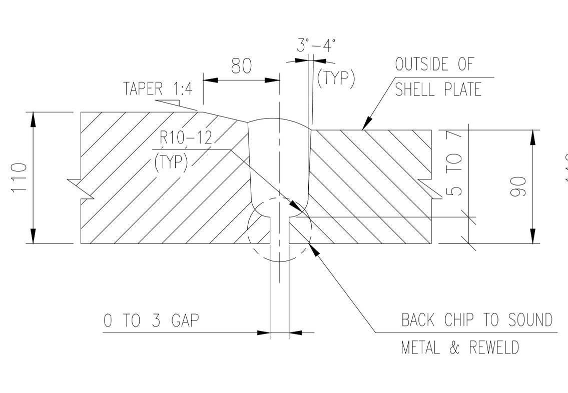
WELD PREPARATION OF LONGITUDINAL SEAM
FOR STEAM DRUM
(SEAM MK. Nos. L-3 TO L-5)



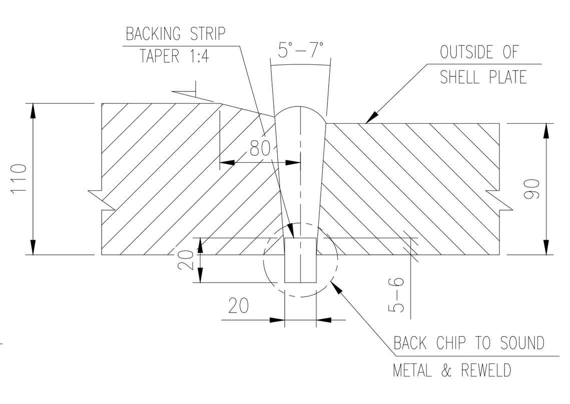
WELD PREPARATION OF CIRCUMFERENTIAL SEAM
(SEAM MK. Nos. C-2 & C-3)



ALTERNATE DETAIL

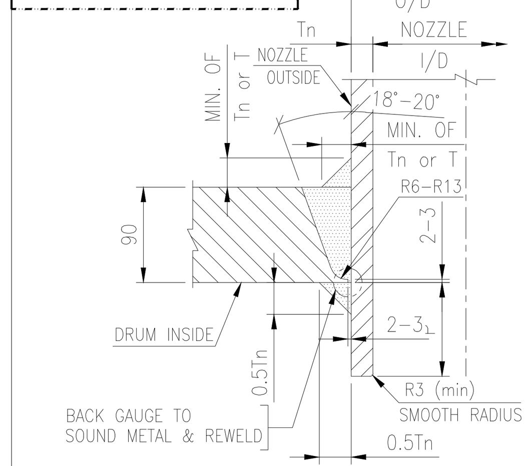


WELD PREPETATION OF CLOSING SEAM
FOR STEAM DRUM MARK Nos. C1 & C4

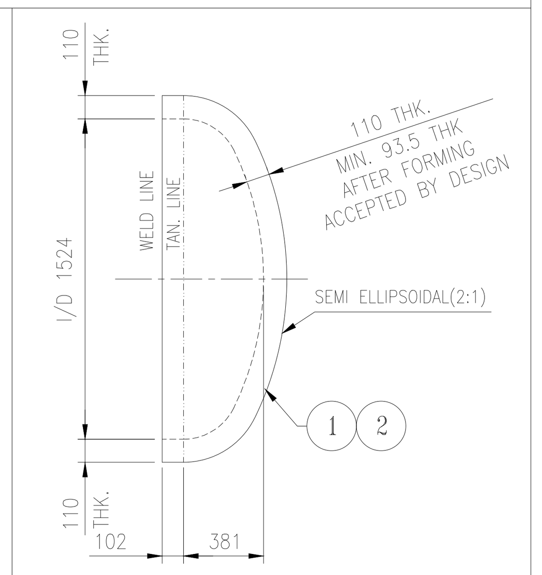


ALTERNATE DETAIL

IBR FIG : XII/85E J1
CHAPTER : XII



WELDING DETAIL OF NOZZLE TO DRUM
(FOR MK.No. N8/1 & N8/2)

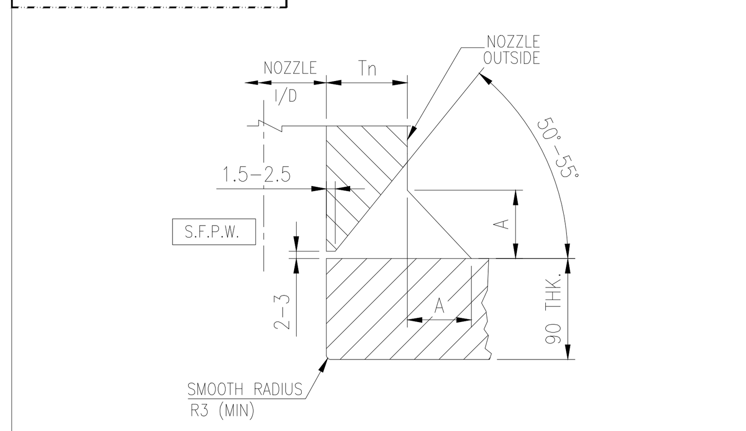


STEAM DRUM DISHED END DETAIL

TABLE - 1

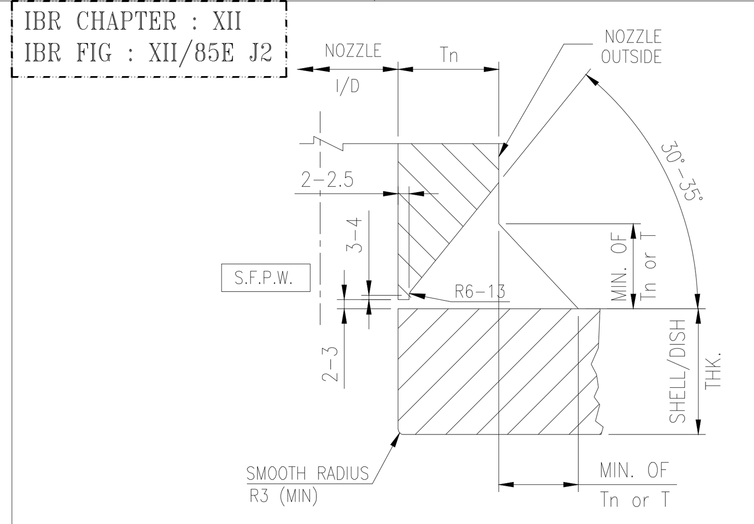
S.Nos.	Nozzle Nos.	Nozzle Descriptions	Location of Nozzle	No. of Nozzle	Nozzle Size (OD) (D) mm	Nozzle Thickness (t) mm	Connecting Pipe Thickness (mm)	Projection from End of Drum (L) mm	Nozzle CL to Drum Vertical Axis (±X) mm	Nozzle CL to Drum Horizontal Axis (±Y) mm	Calculated Nozzle Length (R) mm	Finished Length of the Nozzle (M) mm	M/C Bore for end Preparation (E) mm	Weld Dim. (A) mm	Projection - In from cl of nozzle (p) mm
1	N1	SAFETY VALVE NOZZLE - 1	STEAM DRUM SHELL	1	135	38.93	-	210	-	-	248	208	57.15	39	-
2	N2	SAFETY VALVE NOZZLE - 2	STEAM DRUM SHELL	1	135	38.93	-	210	-	-	248	208	57.15	39	-
3	N3/1 TO N3/4	SATURATED STEAM STUB	STEAM DRUM SHELL	4	200	51.41	8.56	210	-	-	250	210	97.18	52	0
4	N4/1 TO N4/6	COMBUSTOR SIDE WALL RISER STUB	STEAM DRUM SHELL	6	260	60	14.27	250	-	-	290	250	139.76	60	0
5	N5/1 TO N5/6	COMBUSTOR FRONT WALL RISER STUB	STEAM DRUM SHELL	6	260	60	14.27	250	-	-	290	250	139.76	60	0
6	N6/1 TO N6/10	SEPARATOR FRONT, REAR & SIDE WALL RISER STUB	STEAM DRUM SHELL	10	200	51.41	8.56	210	-	-	250	210	97.18	52	0
7	N7	FEED INLET NOZZLE WITH THERMAL SLEEVE	STEAM DRUM SHELL	1	168.3	21.95	14.27	455	-	-	REF. DRG. No.009-0001				125
8	N8/1 & N8/2	DOWN COMER NOZZLE	STEAM DRUM SHELL	2	426	72.48	21.44	210	-	-	440	400	281.04	REF. DRG. No. 009-0001	100
9	N9	CHEMICAL INJECTION NOZZLE WITH THERMAL SLEEVE	STEAM DRUM DISHED END	1	60.3	11.07	8.74	-	165	450	REF. DRG. No.009-0001				60
10	N10/1 & N10/2	AIR VENT NOZZLE	STEAM DRUM SHELL	2	48.3	10.15	5.08	210	-	-	250	210	38.14	10	0
11	N11/1 & N11/2	PR.TRANSMITTER NOZZLE	STEAM DRUM SHELL	2	33.4	9.09	4.55	210	-	-	250	210	24.30	10	0
12	N12/1 & N12/2	PR. GAUGE NOZZLE	STEAM DRUM SHELL	2	33.4	9.09	4.55	210	-	-	250	210	24.30	10	0
13	N13	INST. TEST GAUGE NOZZLE	STEAM DRUM SHELL	1	33.4	9.09	4.55	210	-	-	250	210	24.30	10	0
14	N14	NITROGEN BLANKTING	STEAM DRUM SHELL	1	33.4	9.09	4.55	210	-	-	250	210	24.30	10	0
15	N15	CBD NOZZLE	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	165	400	630	590	24.30	10	212
16	N16	IBD NOZZLE	STEAM DRUM SHELL	1	48.3	10.15	5.08	210	-	-	250	210	38.14	10	0
17	N17	LEVEL TRANSMITTER (STEAM)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350	533	383	343	24.30	10	0
18	N18	LEVEL TRANSMITTER (WATER)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350	533	383	343	24.30	10	0
19	N19	LEVEL TRANSMITTER (STEAM)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350	533	383	343	24.30	10	0
20	N20	LEVEL TRANSMITTER (WATER)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350	533	383	343	24.30	10	0
21	N21	LEVEL TRANSMITTER (STEAM)	STEAM DRUM SHELL	1	33.4	9.09	4.55	-	-	533	445	405	24.30	10	0
22	N22	LEVEL TRANSMITTER (WATER)	STEAM DRUM SHELL	1	33.4	9.09	4.55	-	-	533	445	405	24.30	10	0
23	N23	LEVEL GAUGE NOZZLE (STEAM)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350	533	383	343	24.30	10	0
24	N24	LEVEL GAUGE NOZZLE (WATER)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350	533	383	343	24.30	10	0
25	N25	LEVEL GAUGE NOZZLE (STEAM)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350	533	383	343	24.30	10	0
26	N26	LEVEL GAUGE NOZZLE (WATER)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350	533	383	343	24.30	10	0
27	N27	E W LEVEL INDICATOR (STEAM)	STEAM DRUM SHELL	1	33.4	9.09	4.55	-	-	533	445	405	24.30	10	0
28	N28	E W LEVEL INDICATOR (WATER)	STEAM DRUM SHELL	1	33.4	9.09	4.55	-	-	533	445	405	24.30	10	0
29	N29	SPARE NOZZLE	STEAM DRUM SHELL	1	33.4	9.09	4.55	210	-	-	250	210	24.30	10	0

IBR CHAPTER : XII
IBR FIG : XII/85E B2



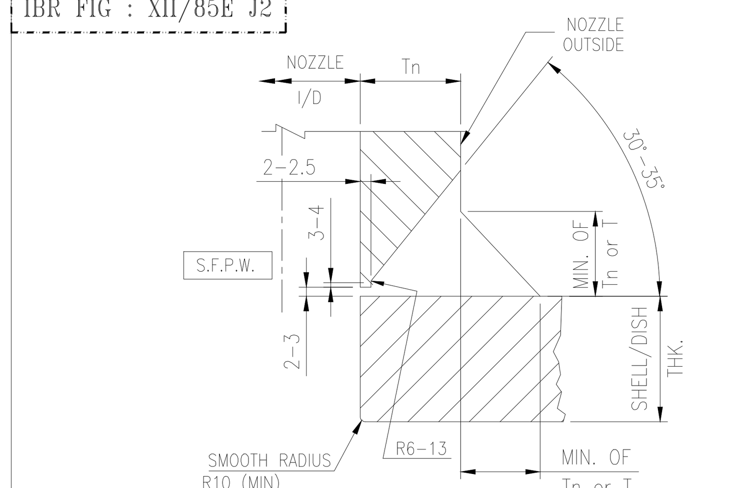
WELDING DETAIL OF NOZZLE TO DRUM (SET ON NOZZLE)
[FOR MARK Nos. N10/1, N10/2, N11/1, N11/2, N12/1, N12/2, N13, N14, N16 TO N29]

IBR CHAPTER : XII
IBR FIG : XII/85E J2



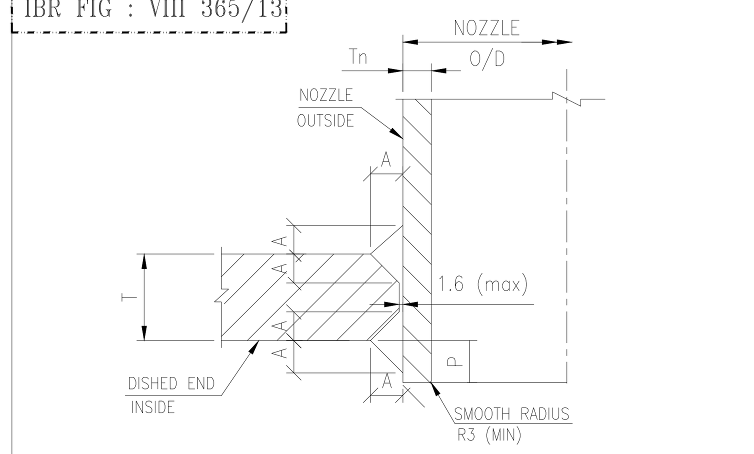
WELDING DETAIL OF NOZZLE TO DRUM (SET ON NOZZLE)
[FOR MK.No. N3/1 TO N3/4, N4/1 TO N4/6, N5/1 TO N5/6 & N6/1 TO N6/10]

IBR CHAPTER : XII
IBR FIG : XII/85E J2

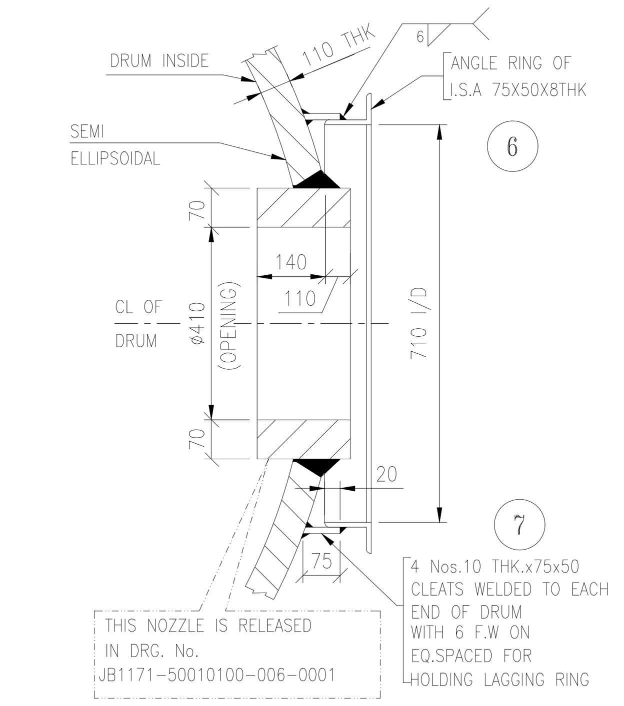


WELDING DETAIL OF NOZZLE TO DRUM (SET ON NOZZLE)
[FOR MARK No. N1 & N2]

IBR CHAPTER : VIII
IBR FIG : VIII 365/13



WELDING DETAIL OF NOZZLE TO DRUM
[FOR MK.No. N15]



DETAIL OF MANHOLE OPENING

SPECIAL NOTES :-

01. STEAM DRUM OF JB1154 IS CHANGED TO JB1171 WITHOUT ANY CHANGE IN PROCESS PARAMETERS. HENCE THE APPLICABLE DRAWING NUMBERS BEARING PROJECT ID JB1154 IS CHANGED TO JB1171. PRESSURE PARTS CALCULATION ARE PREPARED ACCORDINGLY.

DESIGN DATA

CODE OF CONSTRUCTION	IBR 1950 & ITS LATEST AMENDMENTS UP TO - AUGUST 2019 (SEE NOTE : 02)
MAXIMUM WORKING PRESSURE	137 kg/cm^2 (g) [135.4 bar(a)]
MAXIMUM WORKING TEMPERATURE	334 °C
WORKING METAL TEMPERATURE	334 °C
HYDRAULIC TEST PRESSURE	205.5 kg/cm^2 (g) [203.1 bar(a)] (AT SHOP & AS PER IBR AT SITE)
RADIOGRAPHY OF LONG. & CIRCUM. SEAM	100%
STRESS RELIEVING	YES & REFER HEAT TREATMENT CYCLE TABLE
EVAPORATION OF BOILER (M.C.R.)	110 MT/HR
TOTAL HEATING SURFACE AREA OF BOILER (EXCLUDING FORCED FLOW ELEMENT)	1930 sq.m (APROX.)
HEATING SURFACE AREA OF FORCE FLOW SECTION (NON STEAMING)	2025 sq.m (APROX.)
FLANGE PARTICULAR UNLESS STATED OTHERWISE	REF. DRG. NO. : JB1171-50010100-009-0001
STEAM PRESSURE AT SUPER HEATER OUTLET	111.2 Kg/Sq. cm (g)
SUPER HEATER OUTLET TEMPERATURE	540+5,-0 °C
WEIGHT OF DRUM (EMPTY)	31980.10 1
INSPECTION	CIB (HARYANA) (SEE NOTE : 03)

MATERIAL SPECIFICATIONS :

DESCRIPTION	SIZE	SPECIFICATION	MIN. TENSILITY (Kg/ Sq.mm)	MIN.CALC. THK. (mm)	REMARKS
SHELL BELTS	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	83.43	FOR STEAM DRUM
DISHED ENDS	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	92.28	FOR STEAM DRUM
MAN HOLE DOOR	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	84.10	FOR STEAM DRUM

GENERAL NOTES :-

01. ALL DIMENSIONS ARE IN 'mm' UNLESS OTHERWISE SPECIFIED.
02. THE ZERO DATE OF THIS CONTRACT IS 12.08.2019. HENCE IBR AMENDMENTS UP TO AUGUST 2019. HAVE BEEN TAKEN CARE OF WHILE DESIGNING THE EQUIPMENT.
03. INSPECTION BY CIB HARYANA BECAUSE MANUFACTURING OF THIS EQUIPMENT SHALL BE DONE AT ISGEC WORKS, YAMUNA NAGAR (HARYANA)
04. CIRCUMFERENTIAL DIMENSION (ON OUTSIDE TRAMMELS ON DRUM)
05. ALL CIRCUMFERENTIAL DIMENSIONS ARE MEASURED OUTSIDE OF DRUM.
06. ALL FILLET WELDS ARE 6mm UNLESS OTHERWISE SPECIFIED.
07. MANHOLE CIRCULAR TYPE SIZE ø410 FOR DRUM REF. DRG. No. JB1171-50010100-006-0001.
08. A MAX. 6mm FILLET WELD TO BE USED FOR WELDING OF CLEAT BRACKETS AS PER DRG.No : 011-0001 TO 014-0001 AFTER STRESS RELIVING.
09. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH THE FOLLOWING DRGS.
   (a) FOR NOZZLE & STUBS DETAILS REFER DRG.No : JB1171-50010100-009-0001
   (b) FOR EXTERNAL AND INTERNAL BAFFLES & PIPE FERER DRG.No : 010-0001 TO 015-0001.
10. ALL WELDING TO BE DONE BY IBR WELDER USING IBR WELDING PROCEDURES
11. **PAINTING NOTES:-**
    FOR PAINTING NOTES REFER SPECIFICATION NO:JB1171-50570900-PSH-0001.
12. ALL THE PADS WELDED ALL-ROUND ON PRESSURE PART SHALL BE PROVIDED WITH TELL-TALE HOLE OF DIA 5 mm ON THE PAD TO RELEASE ENTRAPPED AIR/GAS.
13. H.W.L - DENOTES HIGH WATER LEVEL, H.H.W.L - DENOTES HIGH HIGH WATER LEVEL, N.W.L - DENOTES NORMAL WATER LEVEL, L.W.L - DENOTES LOW WATER LEVEL, L.L.W.L - DENOTES LOW LOW WATER LEVEL.
14. DRUM TO BE STAMPED AS FOLLOWS :-

ITEM CODE-SDM001-STEAM DRUM
ISGEC HEAVY ENGINEERING LTD
WORKS No.-PD- 0328 (S)/I.W.T.NO.-6655
YEAR OF MAKE-
TESTED TO-205.5 Kg/Sq. cm (g)
TESTED ON-
MAXIMUM ALLOWABLE WORKING PRESSURE-137 Kg/Sq.cm(g)
COMPETENT PERSON'S / INSPECTING AUTHORITY'S OFFICIAL STAMP
FINAL DATE OF EXAMINATION

15. FOLLOWING ADDITIONAL TESTING SHALL BE CARRIED OUT FOR STEAM DRUM.
   (a) ULTRASONIC EXAMINATION OF ALL BUTT WELD AFTER PWHT SHELL BE PERFORMED AT SHOP INCASE OF PL THK>=100 mm.
   (b) MAGNETIC PARTICLE EXAMINATION / DIE PENETRATION TEST OF NOZZLE / STUBS OF SHELL WELDING.
16. FOR DIMENSIONAL TOLERANCE REQUIREMENTS REF. DOC. NO. GEN-50990100-DSM-002 AND ITS ASSOCIATED DRAWINGS.

PWHT CYCLE :- (FOR STEAM DRUM)

A)	LOADING TEMP. (MAX.)	= 300°C
B)	RATE OF HEATING ABOVE 300°C	= 55°C/Hr.
C)	SOAKING TEMP.	= 610°±10°C
D)	SOAKING TIME (MIN.)	= 2Hr.45 MINUTES
E)	RATE OF COOLING UP TO 300°C	= 55°C/Hr.
F)	UNLOADING TEMP.	= 300°C
G)	COOLING AFTER UNLOADING	= COOLING IN STILL AIR

P.NO.	PART DESCRIPTION	SPECIFICATION	SIZE	P.QTY	P.WT(kg)	REMARK
--	FLATS & BRACKETS ON	--	--	1 SET	24.81	JB1171-50010100-013-0001
--	FLATS & STUDS INSIDE	--	--	1 SET	194.60	JB1171-50010100-012-0001
--	DETAIL OF DRUM NOZZLE	--	--	1	1761.52	JB1171-50010100-009-0001
--	A&D OF MAN HOLE DOOR	--	--	2	949.18	JB1171-50010100-006-0001
7	CLEAT FOR RING	IS 2062 E250 A	PL.10 THK-50 X 75	8	2.35	-
6	LAGGING RING	IS 2062 E250 A	ISA 75X50X8 THK - 2576 LG.	2	38.12	-
5	SHELL	SA 516, Gr.70	PL.90 THK-2500 X 5440	1	8809.96	(9608.4 Kg.-MAT.FORECAST WT.)
4	SHELL	SA 516, Gr.70	PL.90 THK-1430 X 5590	1	5008.62	(5647.55 Kg.-MAT.FORECAST WT.)
3	SHELL	SA 516, Gr.70	PL.90 THK-2500 X 5440	1	8805.41	(9608.4 Kg.-MAT.FORECAST WT.)
2	DISHED END	SA 516, Gr.70	PL.110 THK-2170 X 2170	1	3192.35	(4066.14 Kg.-MAT.FORECAST WT.)
1	DISHED END	SA 516, Gr.70	PL.110 THK-2170 X 2170	1	3193.18	(4066.14 Kg.-MAT.FORECAST WT.)

ITEM CODE	ITEM DESCRIPTION	I.TYPE	I.GROUP	I.QTY	I.WT (Kg) (TOTAL)	REMARK
SDM001	STEAM DRUM WITH ATTACHMENTS	P	B215	1	31980.10	-

Issued to _______ Issued by _______ Issued date _______
☐ For pre-planning ☐ For manufacture in shop _______
☐ For provisional manufacture in shop _______ ☐ For sub contracting / site work
This drawing supersedes drawing no. ___________ Rev. ___

W.O. No. PD-0328 / I.W.T. No. :6655

CONSULTANT	M/s. AVANT-GARDE ENGINEERS AND CONSULTANTS (P) LTD.
CLIENT	M/s. GIRIDHAN METAL PRIVATE LIMITED..

DO NOT SCALE
IF IN DOUBT PLEASE ASK

FOR APPROVAL	NO	FOR INFORMATION	NO	FOR PRODUCTION	YES	FOR ERECTION	YES

RESPONSIBLE DEPT.	DRAWN BY	CHECKED BY	APPROVED BY	DATE
MECHANICAL	VENKAT.P	B.SARAVANAN/Y.SREEKANTH	R.ARUN	23.03.2020

TOTAL WEIGHT IN kg.	PROJECT ID	ELEMENT ID	I.W.T. NO.
31760.69	JB1171	50010100	6655
	YEAR : 2019	SHEET SIZE : A1	SCALE :1 : 30

REVN.	ZONE	DESCRIPTION	REVD. BY	CHKD. BY	APPD. BY	DATE
1	~	01. HOLD LIFTED CORRESPONING CHANGES ARE MADE.	V.KAVASKAR	R.G.AMIR/R.KUMARAN	R.ARUNKUMAR	22.06.2020

ISGEC BOILERS
(formerly Isgec John Thompson)
Isgec Heavy Engineering Ltd.
Noida 201 301 (U.P.) India

SERVICE	SINGLE DRUM, TOP SUPPORTED,CFBC BOILER
TITLE	ARRANGEMENT & DETAIL OF STEAM DRUM
DRAWING DOCUMENT ID	JB1171-50010100-007-0001
REVISION	01

THIS IS A CAD FILE - DO NOT CORRECT IT MANUALLY

CAD FILE FROM SOFTWARE : CAD